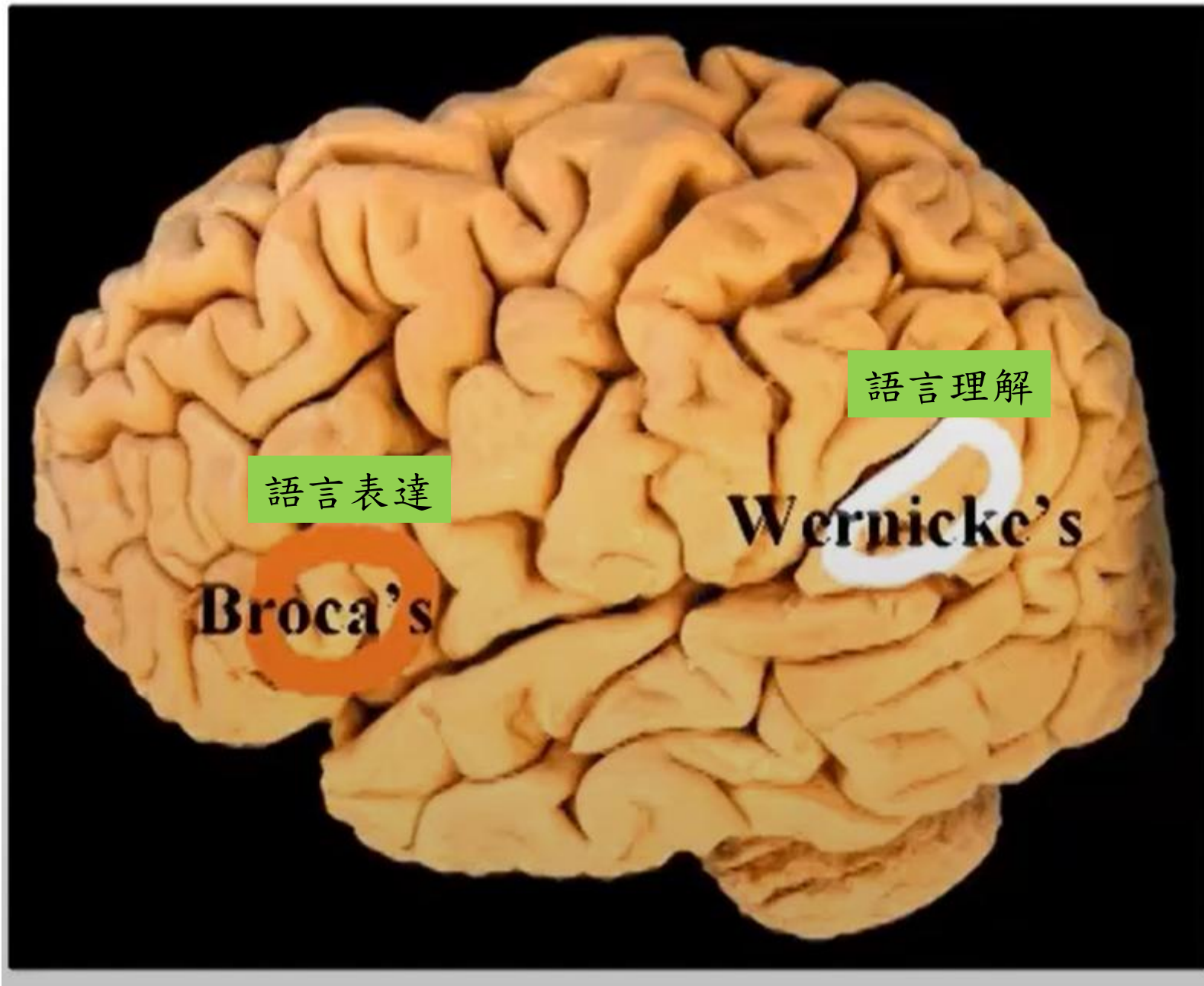


計算認知神經科學 腦與感官





Example of confounding

“Clever” Hans

CLEVER HANS

(THE HORSE OF MR. VON OSTEN)

*A CONTRIBUTION TO EXPERIMENTAL
ANIMAL AND HUMAN
PSYCHOLOGY*

BY
OSKAR PFUNGST

WITH AN INTRODUCTION BY PROF. C. STUMPF,
AND ONE ILLUSTRATION AND FIFTEEN FIGURES

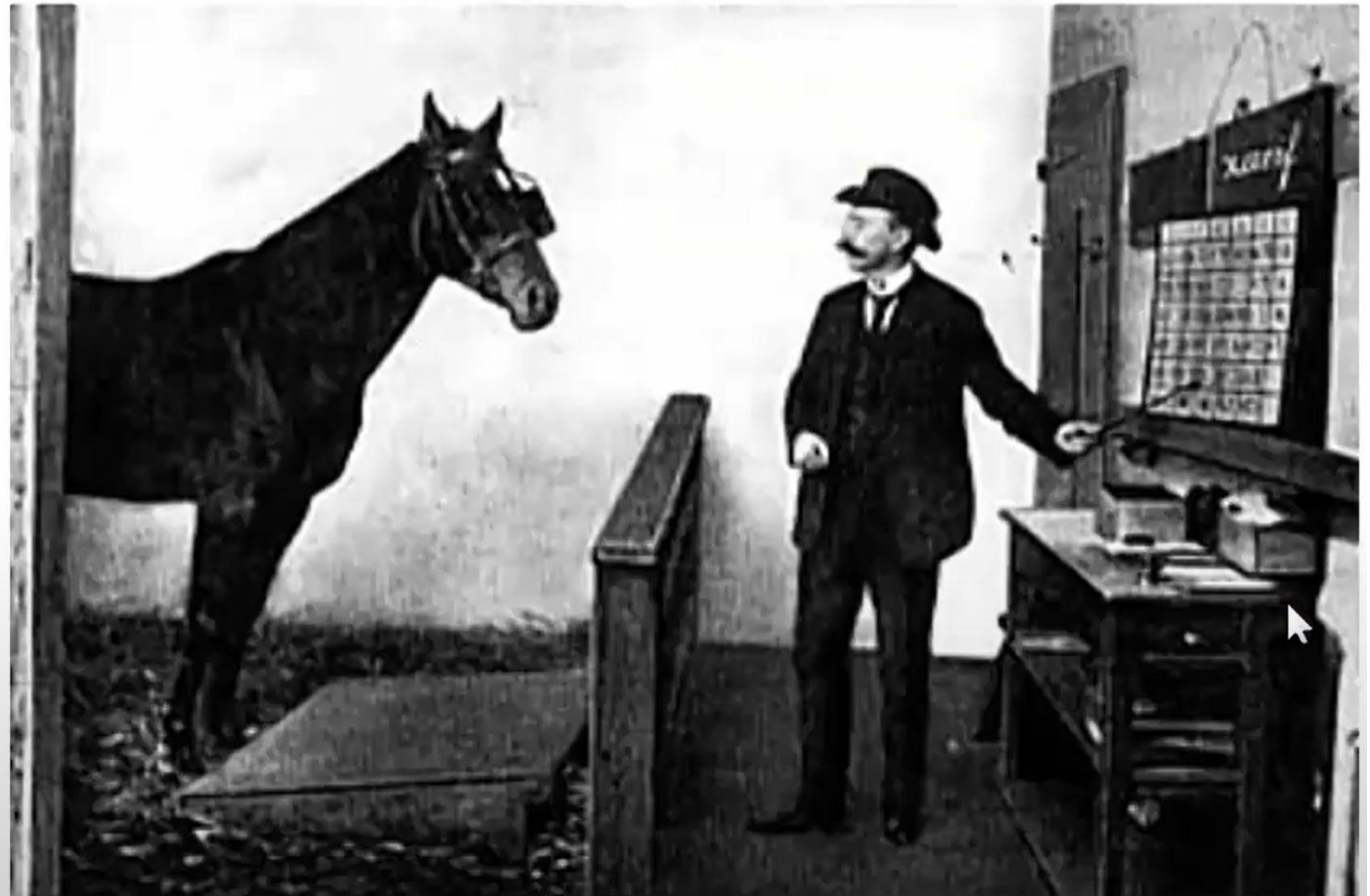
TRANSLATED FROM THE GERMAN
BY
CARL L. RAHN
Professor in Psychology in the University of Chicago

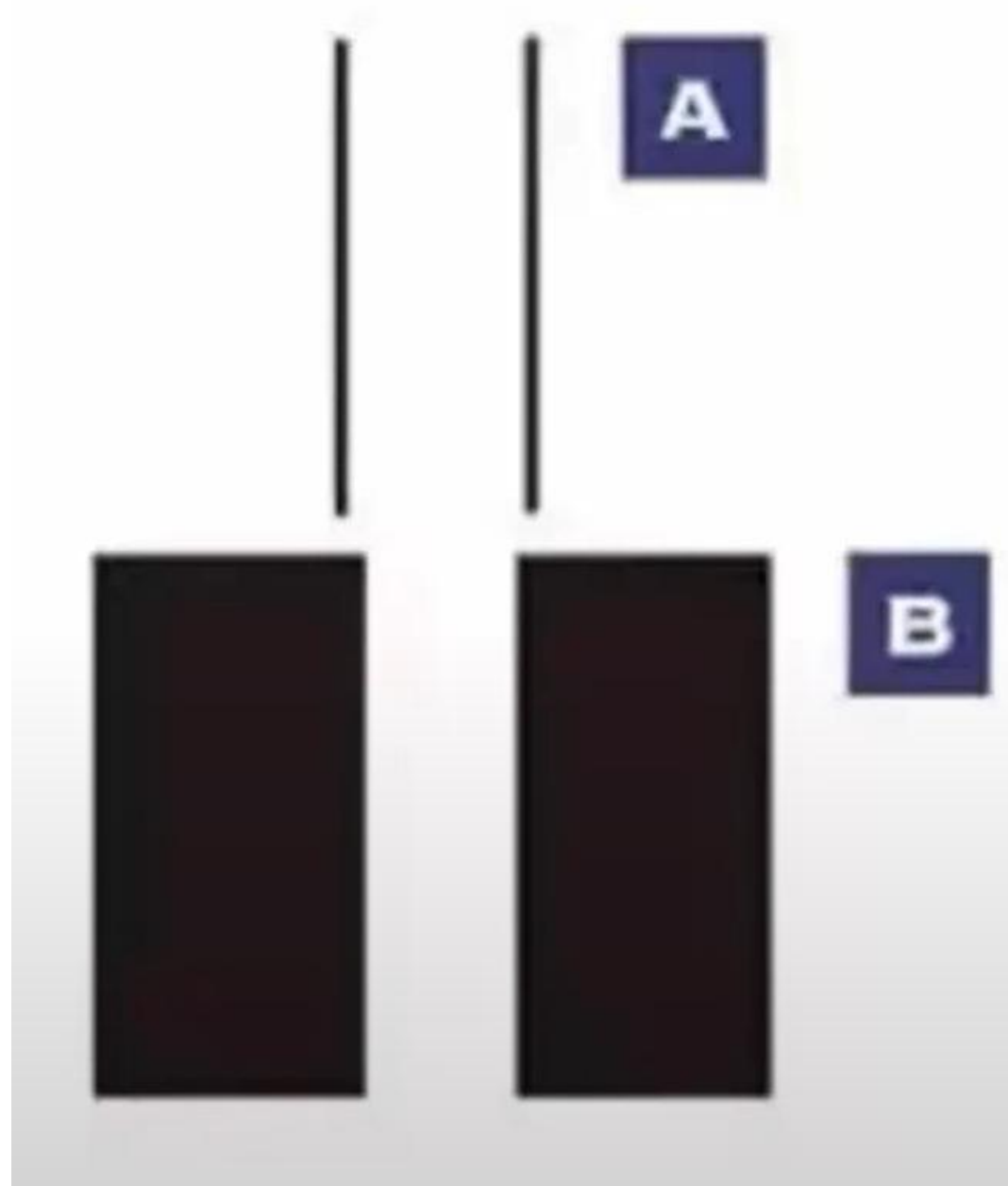
WITH A PREFACE BY
JAMES H. ANGELL
Professor of Psychology in the University of Chicago

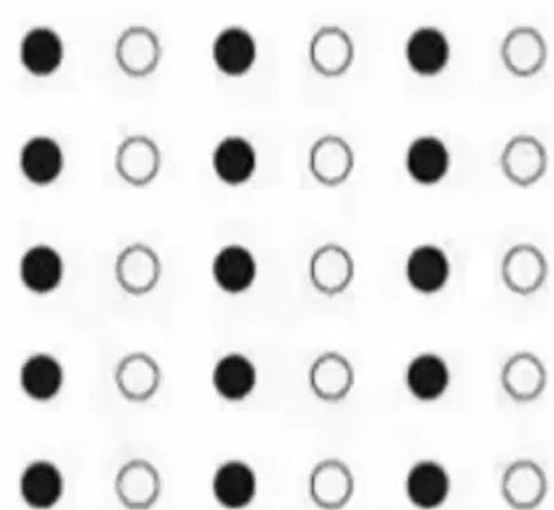
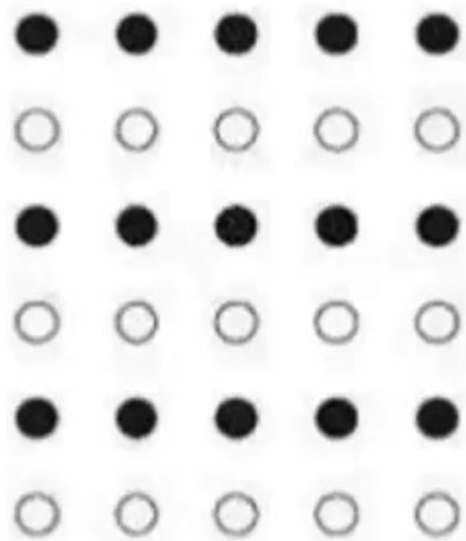


NEW YORK
HENRY HOLT AND COMPANY

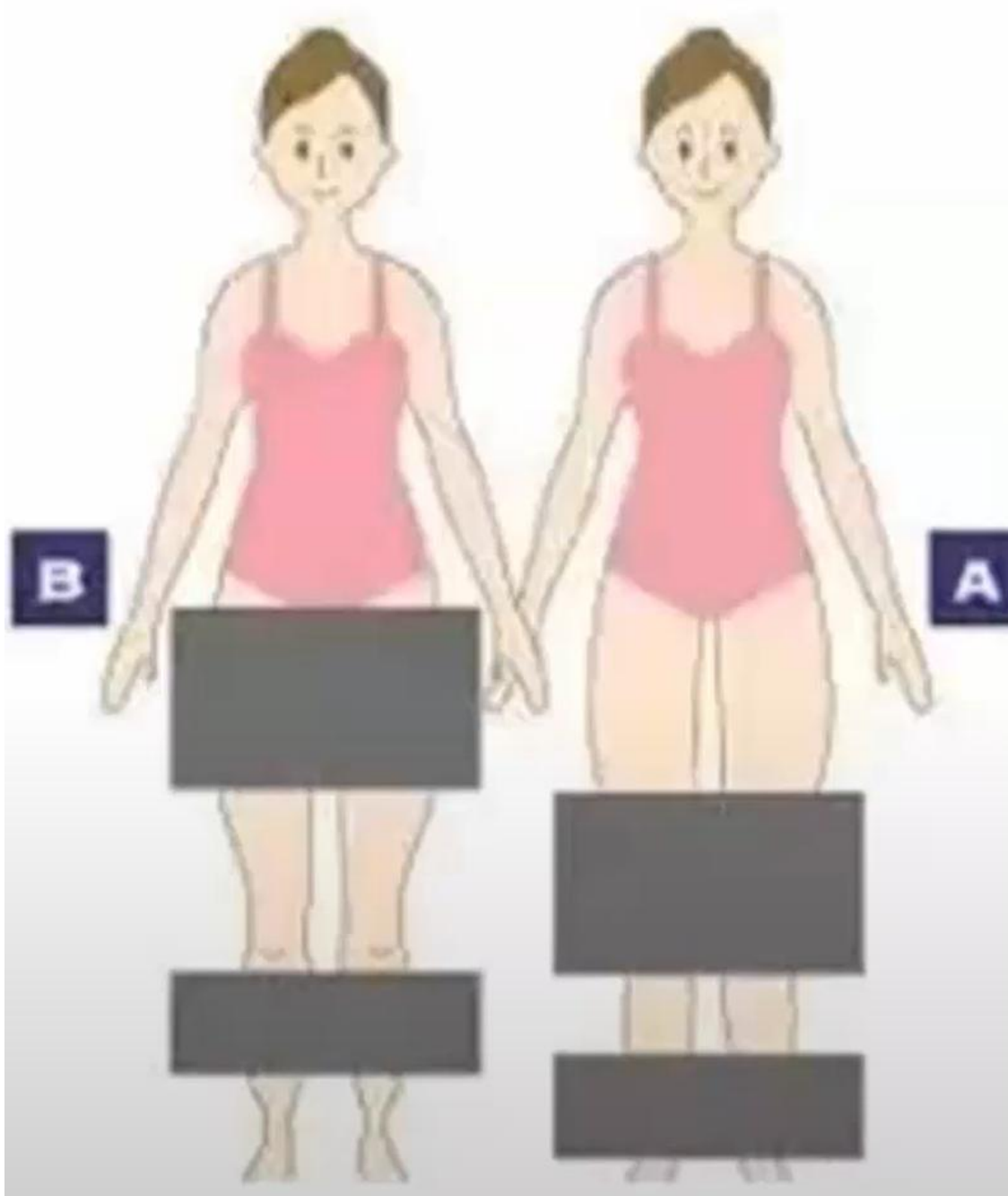
1902

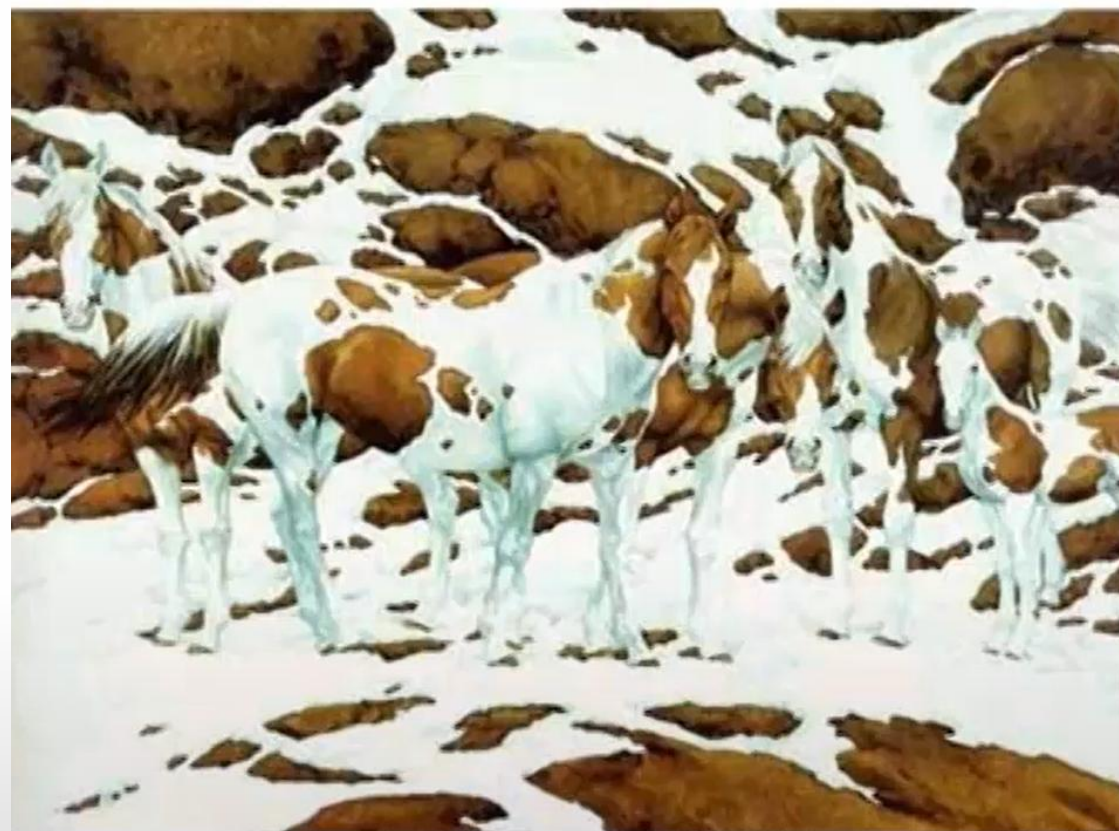
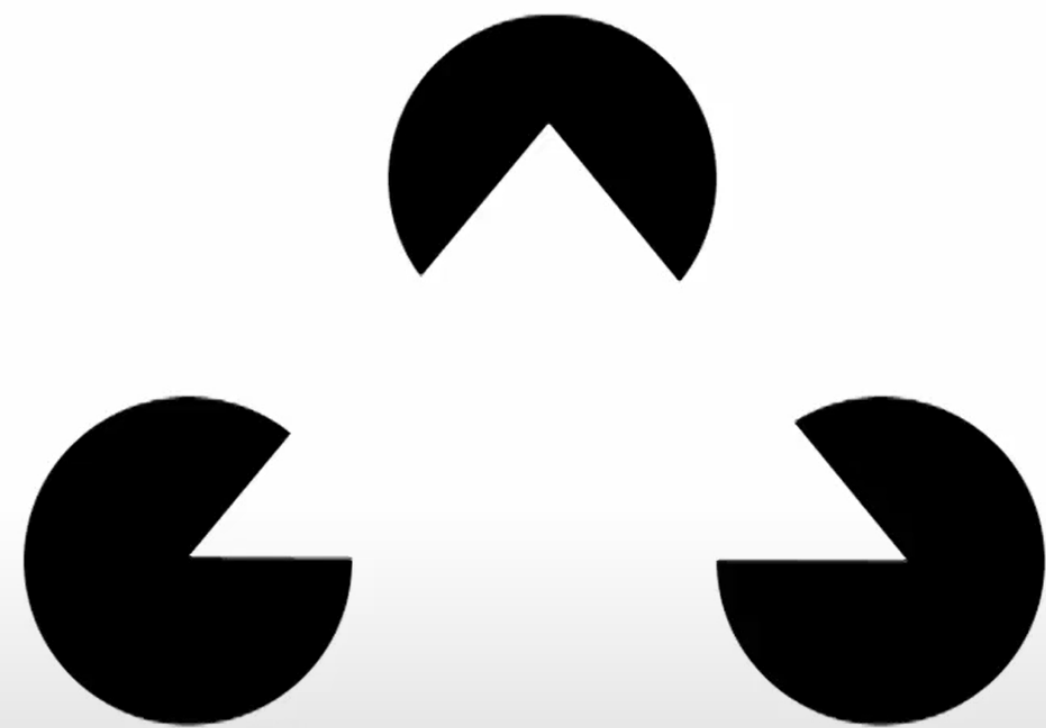








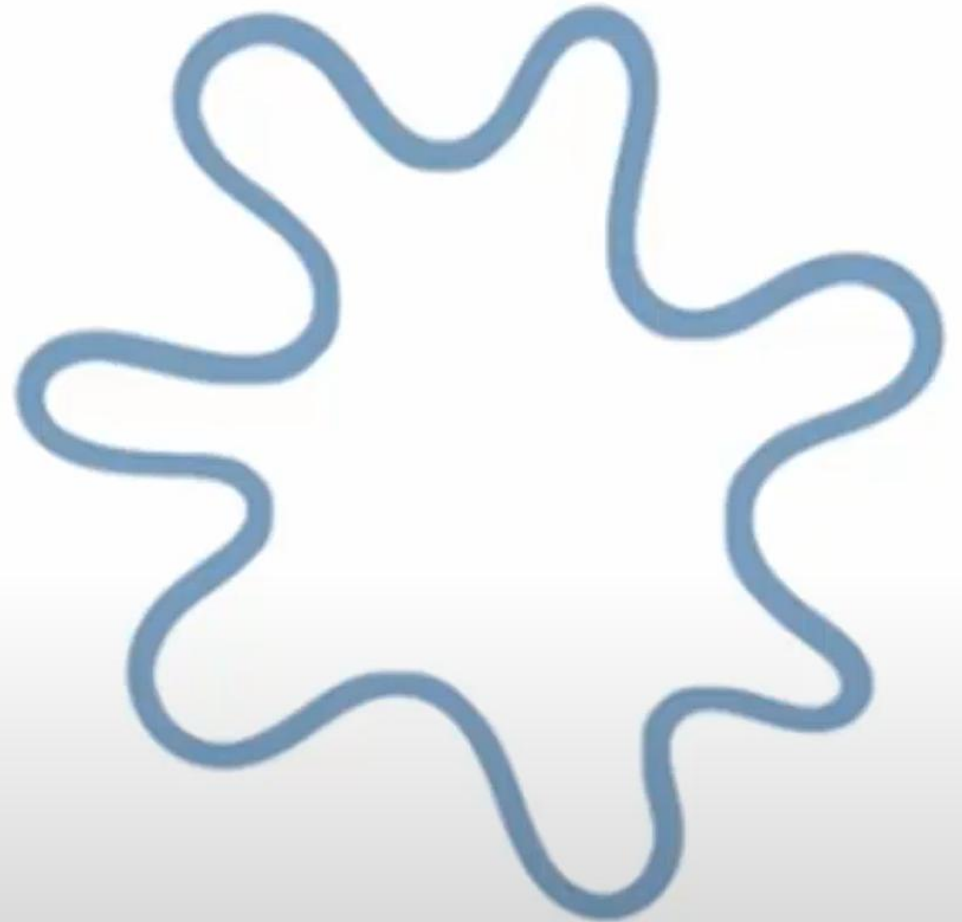




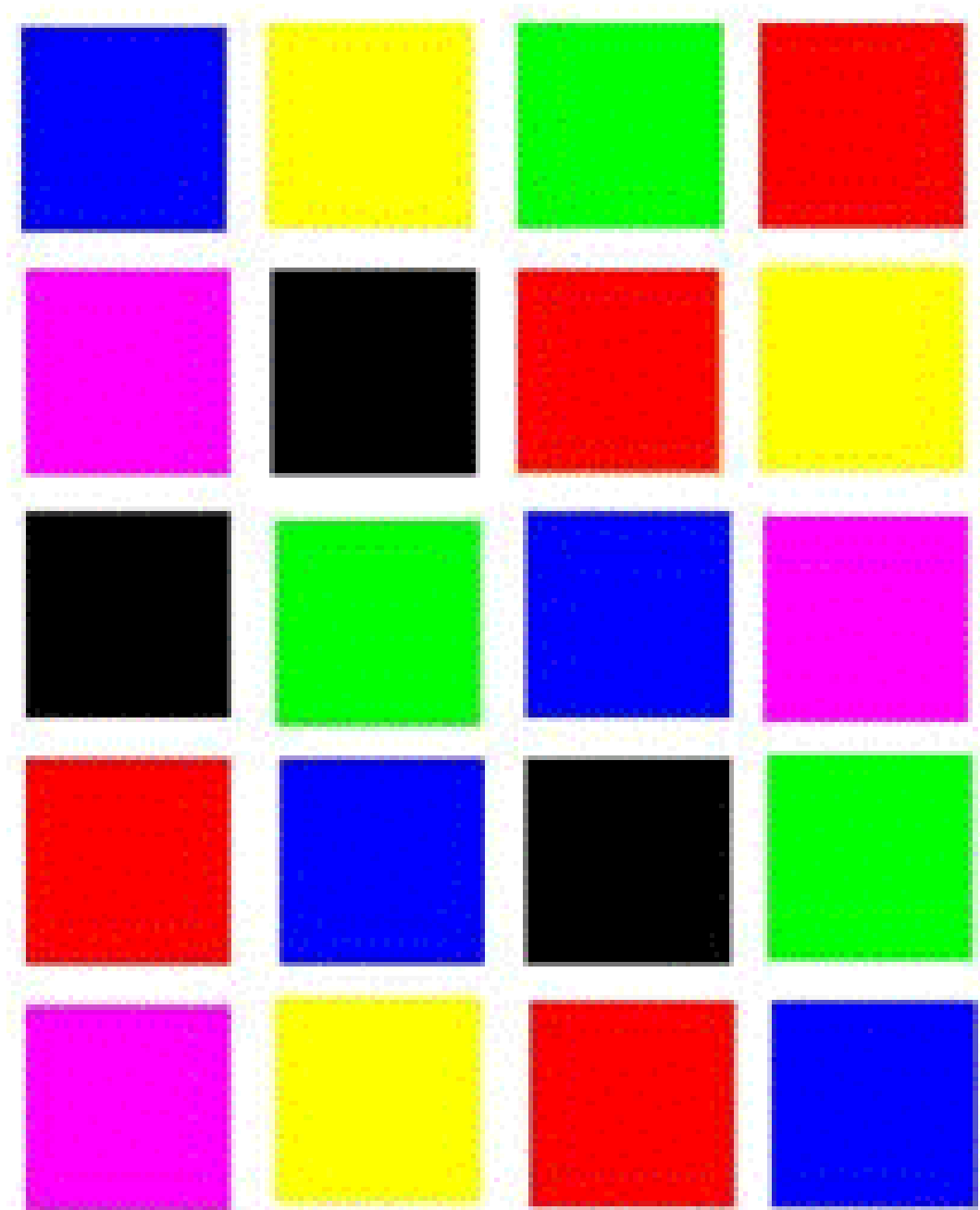
Are face special?



Who is Kiki? Who is Bobua?



史楚普叫色實驗



史楚普叫色實驗



Psychological experiment: Color-word Stroop task

- Slower response for incongruent trials (the color of a word is not the meaning of the word).
- Semantic process is automatic.
- You need to actively suppress the information processing of semantic knowledge.

紅

Congruent trial

紅

Incongruent trial

Topics in cognitive neuroscience

(incomplete list)

- Sensation & Perception
- Object Recognition
- Control of Action
- Learning and Memory
- Emotion
- Attention & Consciousness
- Language comprehension & Production
- Thinking & Decision Making
- Plasticity
- Social cognition
- And so on...

Representations (表徵)

- Definition: **physical properties of the world** (e.g., sounds, colors) and **abstract forms of knowledge** (e.g. knowledge of the beliefs of other people, factual knowledge).
- Representations are manifested in:
 - Cognitive system:
 mental representation
 (cognitive psychology)
 - Neural system:
 neural representation
 (neuroscience)

Representations (表徵)

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Representations (表徵)

- Cognitive system:
 mental representation
 - Mental model of the outside world
 - ✓ e.g., colors, objects, names



Universal design

the applications of mental representation



http://a-chien.blogspot.tw/2012/11/blog-post_4.html

Universal design

the applications of mental representation

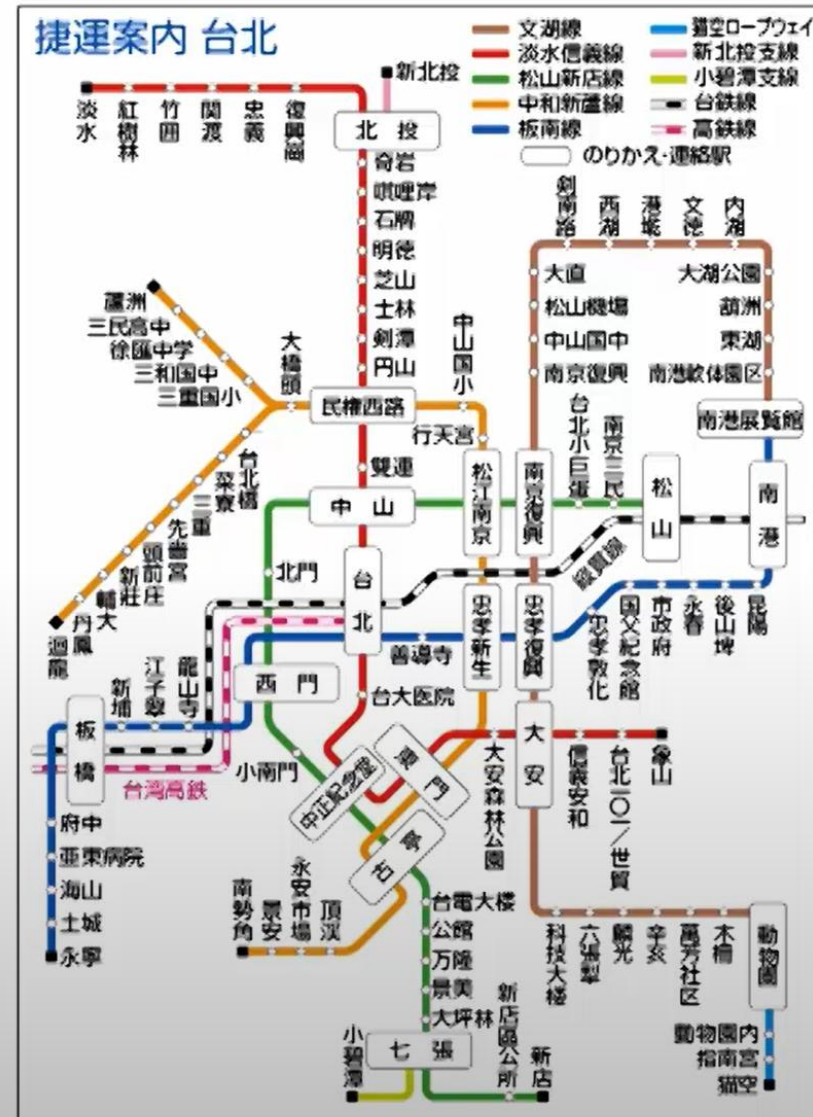


Representations is NOT identical

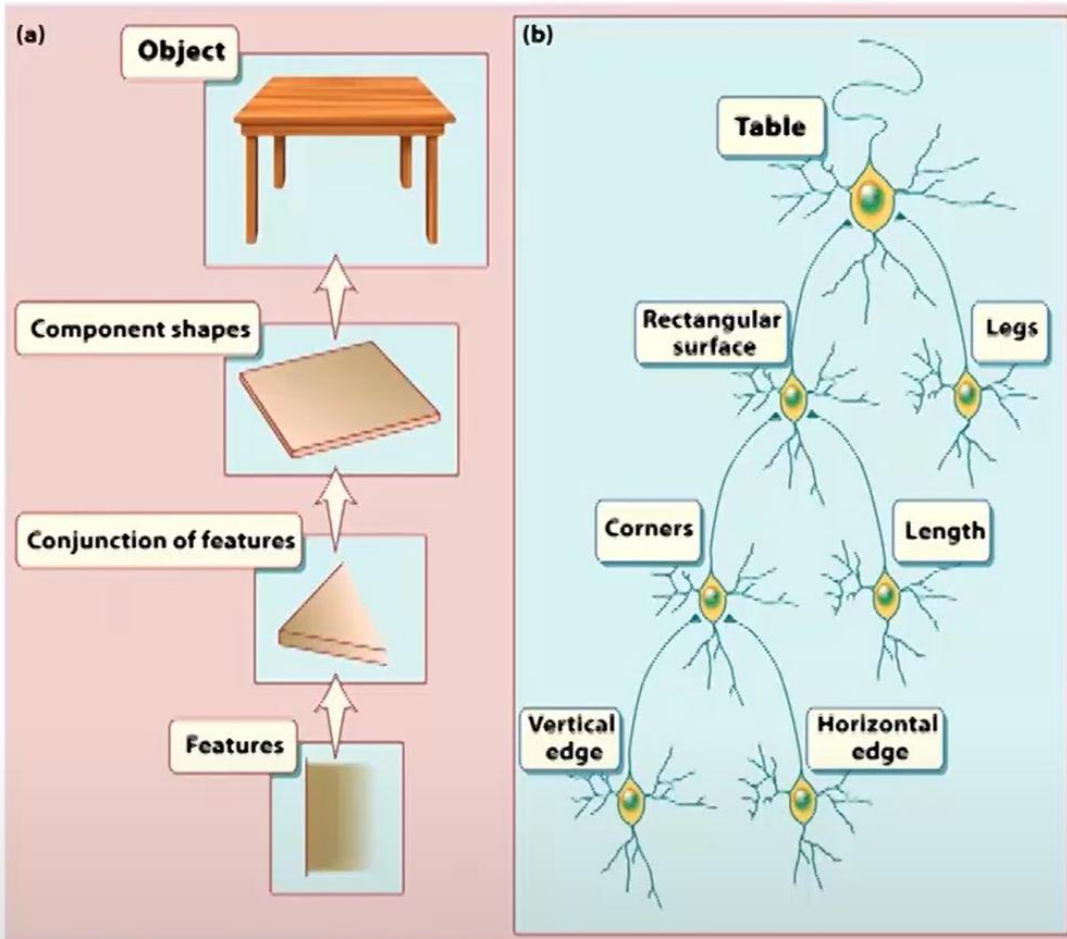
Taipei metro map in Taiwan



Taipei metro map in Japan



Representations

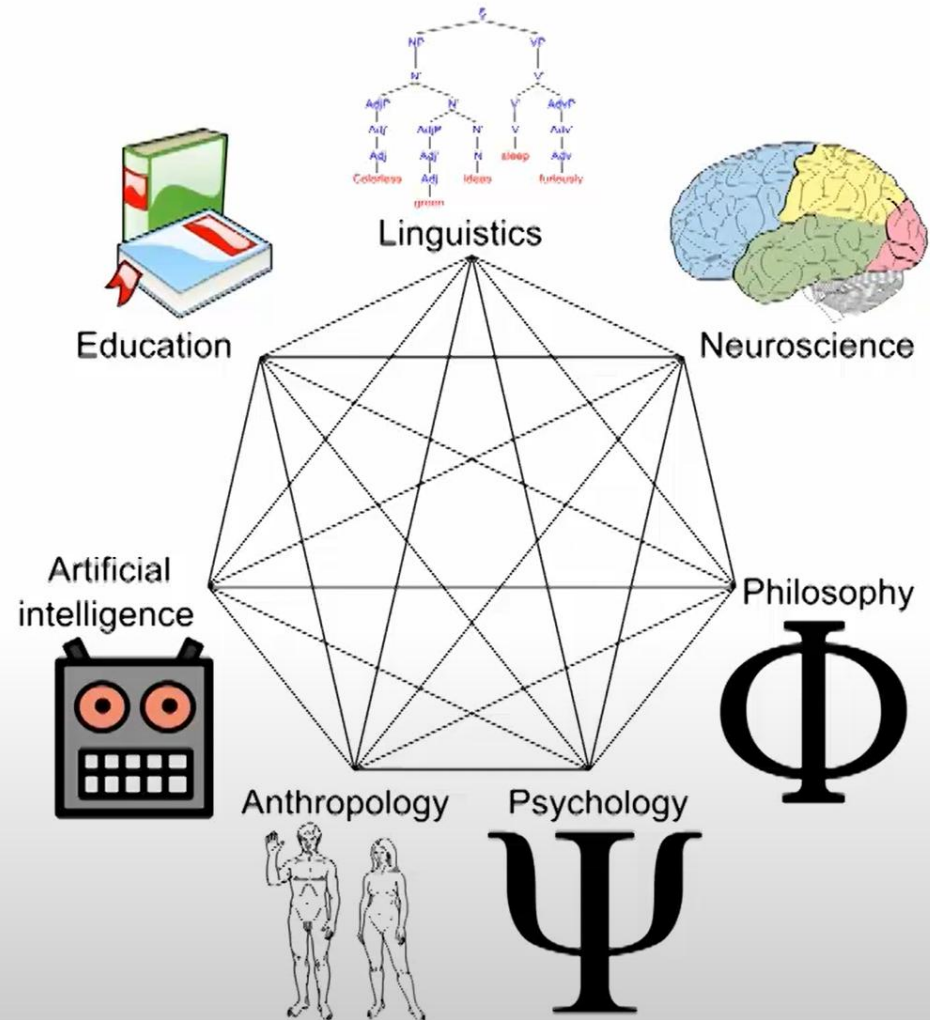
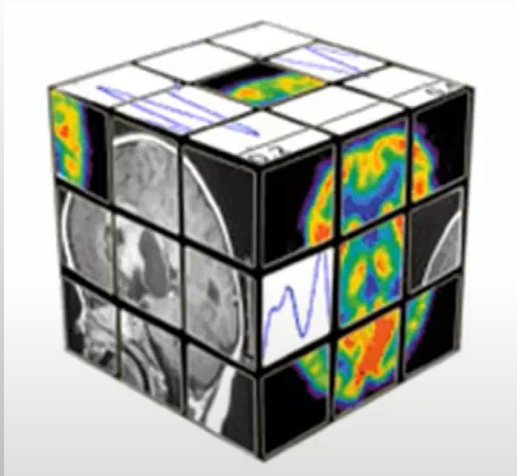


- Neural system
 - Neural representation
 - The properties of the outside world manifest themselves in the neural signal / biological system
 - e.g., edges, rectangle



What is **cognitive science**?

- Interdisciplinary scientific study of the mind and its processes
- Examines:
 - what cognition is
 - what cognition does
 - how cognition works



<https://apps.carleton.edu/curricular/cgsc/>



Gustatory

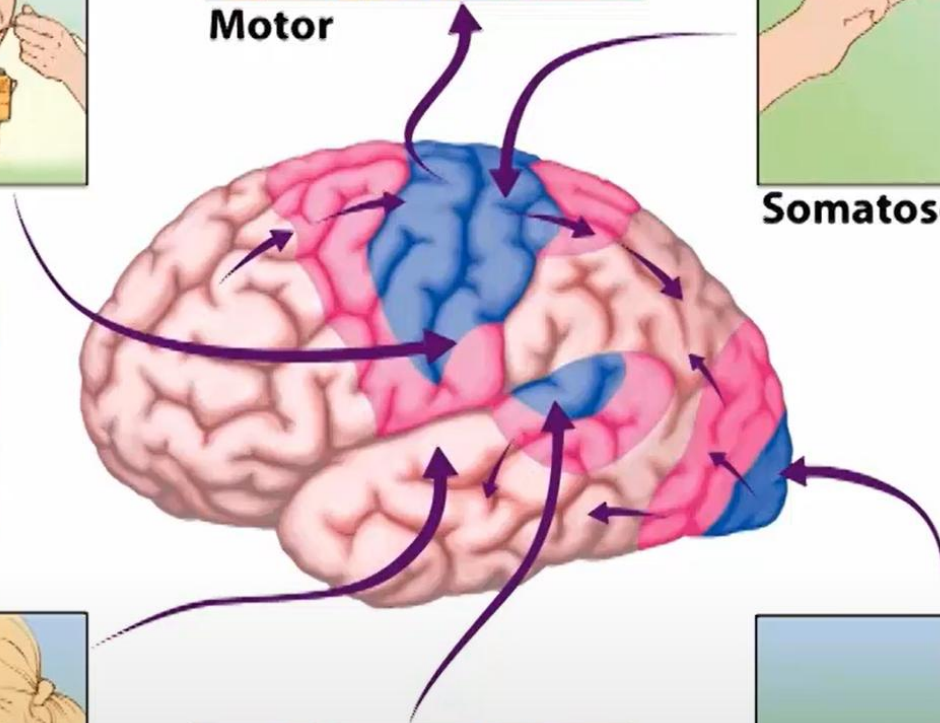


Motor



Somatosensory

- Primary sensory or motor cortex 
- Secondary sensory or motor cortex 
- Association cortex 



Olfactory



Auditory



Visual

What is cognitive neuroscience (Brain and Cognitive Science)?

Cognitive Neuroscience is the marriage of **neuroscience** (studying the underlying brain mechanisms) and **cognitive psychology** (studying cognition at the behavioral level).

➤ **Neuroscience:**

- ✓ Varieties of Species
- ✓ Level of Analysis

➤ **Cognitive Psychology**

- ✓ Studying cognition at the behavioral level
- ✓ mental processes: thinking, perceiving, imaging, speaking, acting, and planning

The basic concept of images

- Before any image processing, we need to understand the “format” of the image.
- Neural images are “digital” images.

low resolution



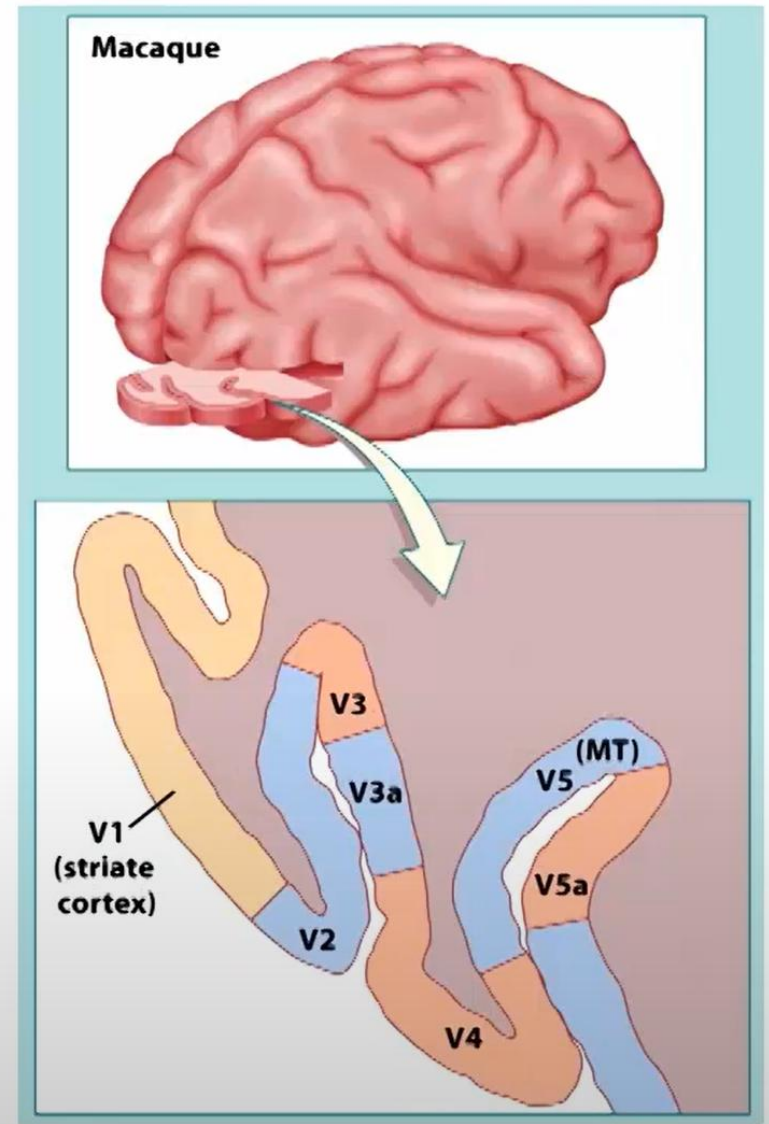
high resolution



Occipital lobe: cortical visual areas

- Map of cortical areas:

- Microelectrodes inserted in cortical areas show many areas responding to visual stimuli (in monkeys)
 - ✓ Labeled V1, V2, V3, V4, etc.
- Neurons in different areas are sensitive to different types of stimuli

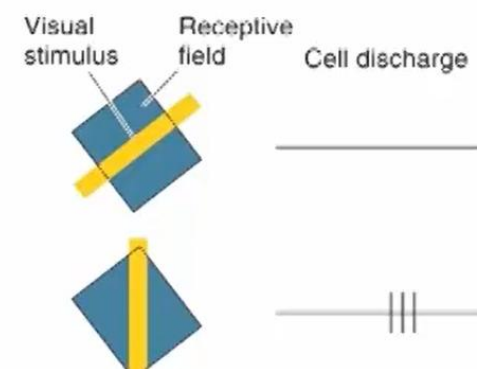
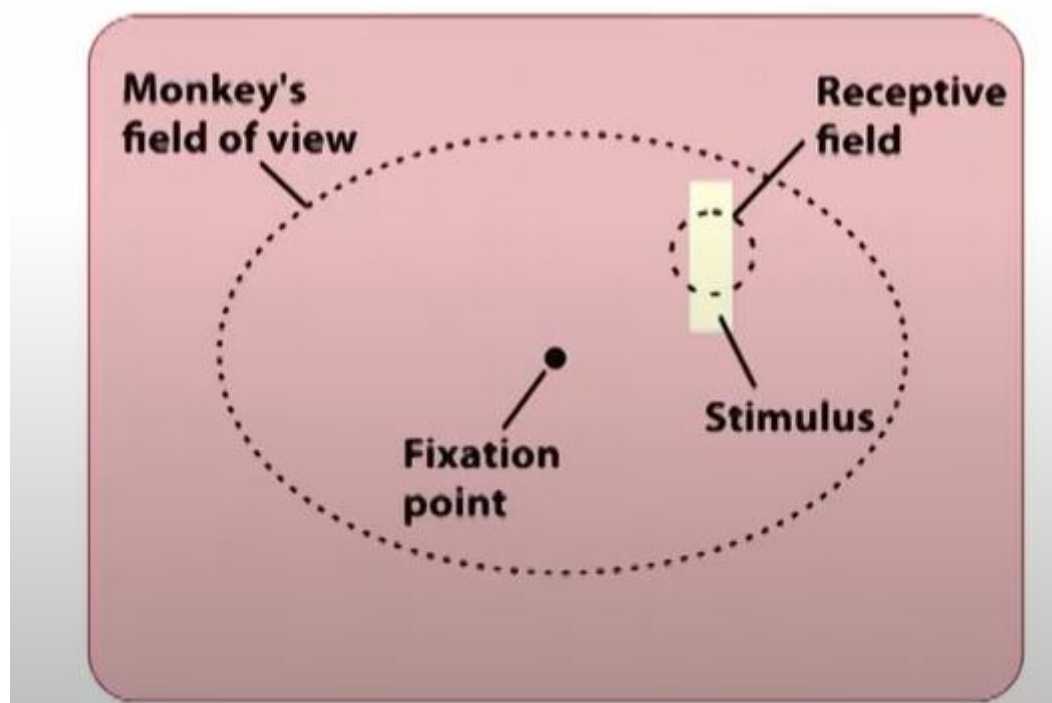


Occipital lobe: cortical visual areas

- A functional specialization of occipital visual areas

➤ V1

- ✓ Optimal stimuli: bars, edges, directions, etc.
- ✓ Partial separation between eyes
- ✓ Receptive fields: small

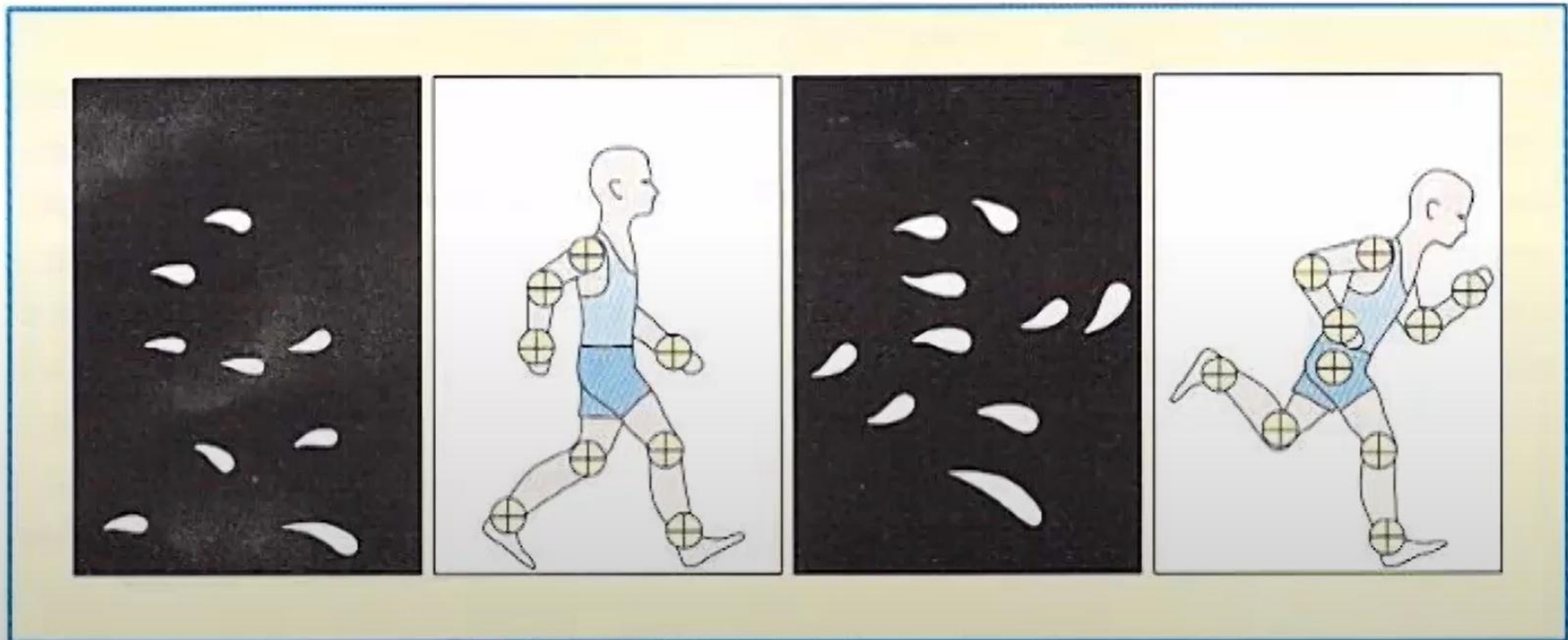


Occipital lobe: cortical visual areas

- A functional specialization of occipital visual areas
 - **V2-V4** (visual areas 2-4)
 - ✓ Optimal stimuli: Same as V1, plus moving bars
 - ✓ Most neurons receive input from both eyes
 - ✓ Receptive fields: progressively larger
 - **V4: color center of the brain**

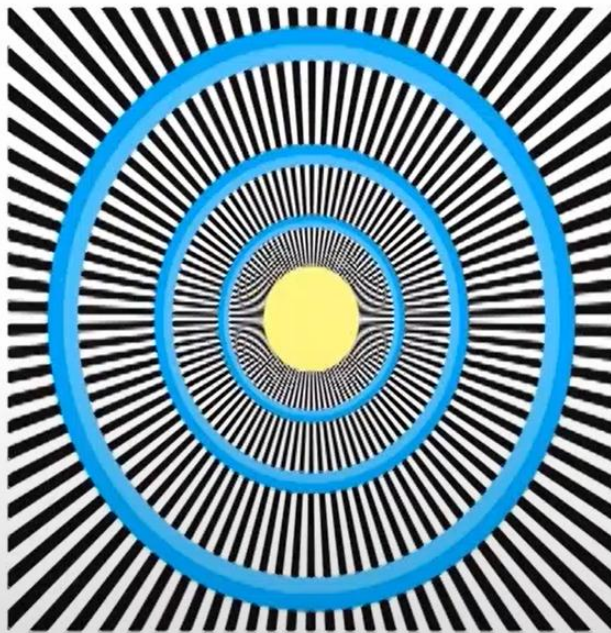
Occipital lobe: cortical visual areas

- A functional specialization of occipital visual areas
 - **V5 (also called MT, border with temporal lobe)**
 - ✓ **Movement center** of the brain
 - ✓ Optimal stimuli: Only sensitive to moving object
 - ✓ Receptive fields: very big

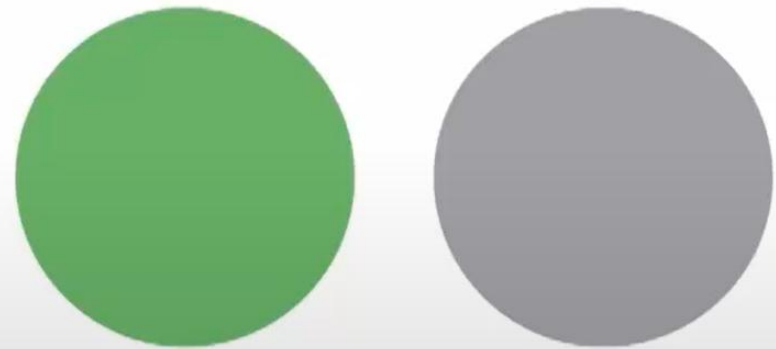


Visual illusions

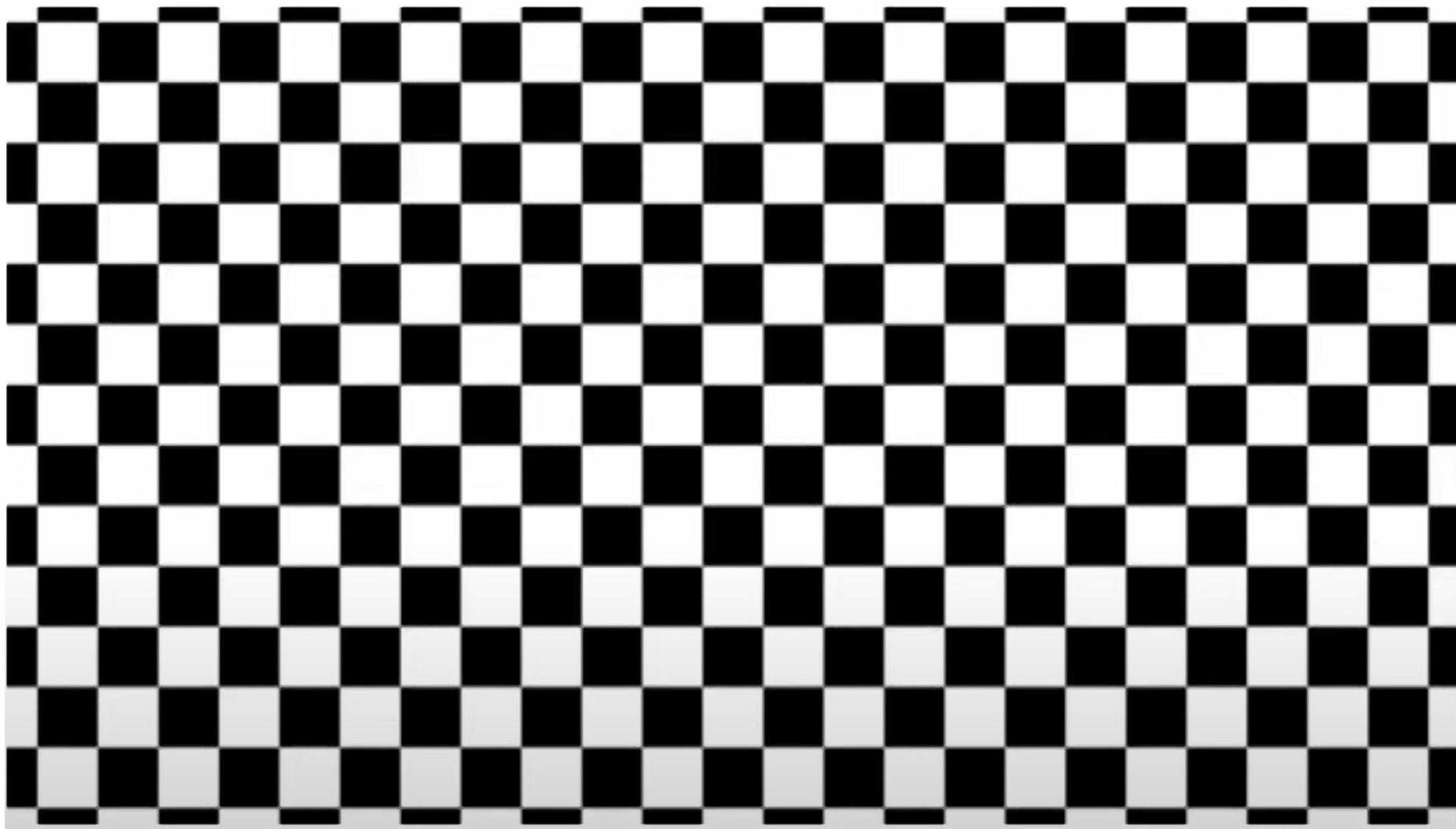
- Our percepts are more closely related to neural activity in higher visual areas.

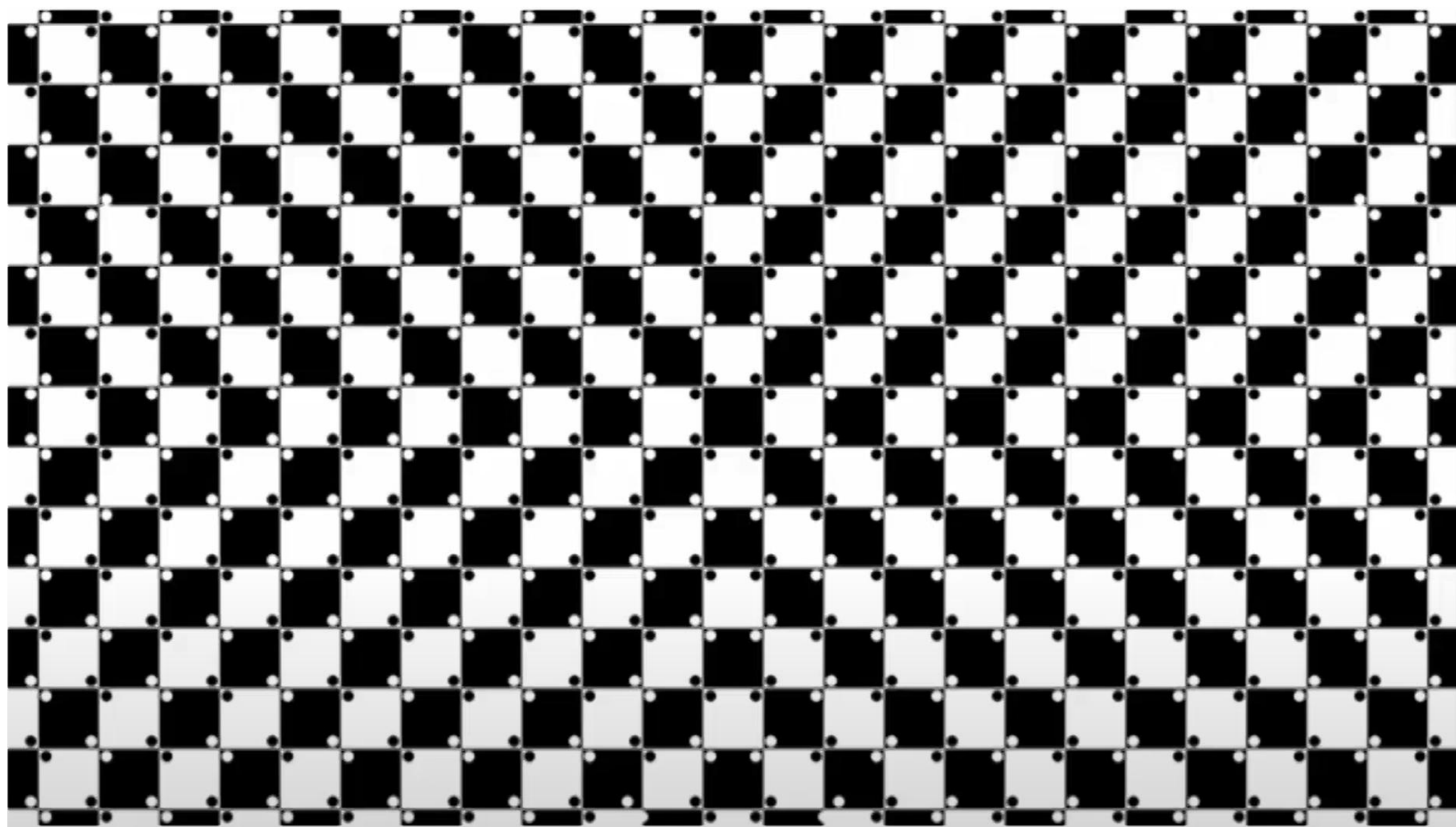


Illusory motion:
activation in **area MT**



Color aftereffects:
activation in **area V4**





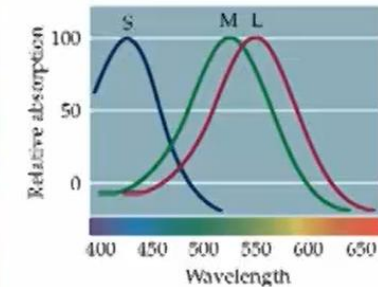
Deficits in visual perception

- Deficits in color perception:
 - Dichromats:
 - ✓ Red-green color-blind: medium/long wavelength missing
 - ✓ Blue-yellow color-blind: short wavelength missing
 - ✓ Normal cortical visual areas

(A) Normal (trichromat)



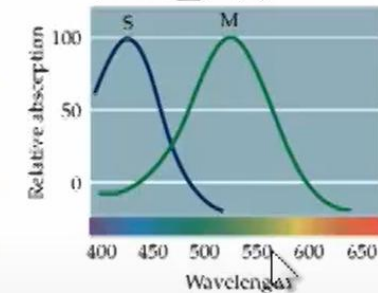
正常視覺



(B) Protanopia



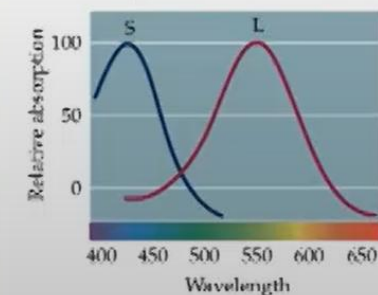
紅色盲



(C) Deuteranopia

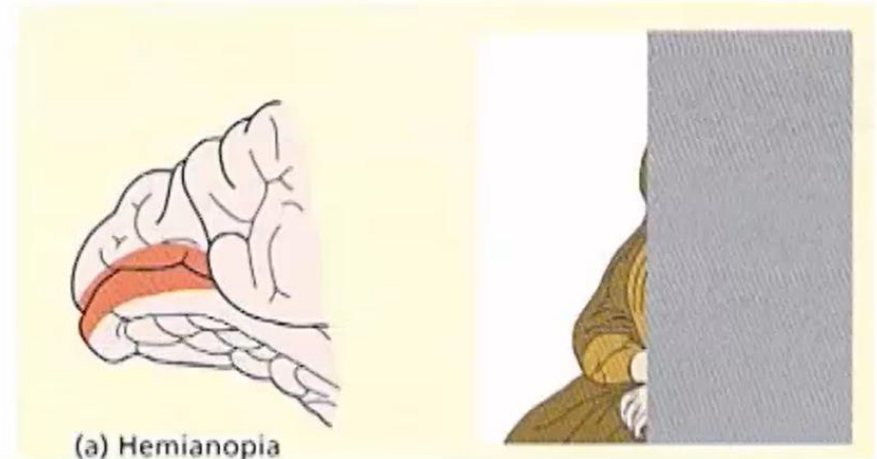


綠色盲



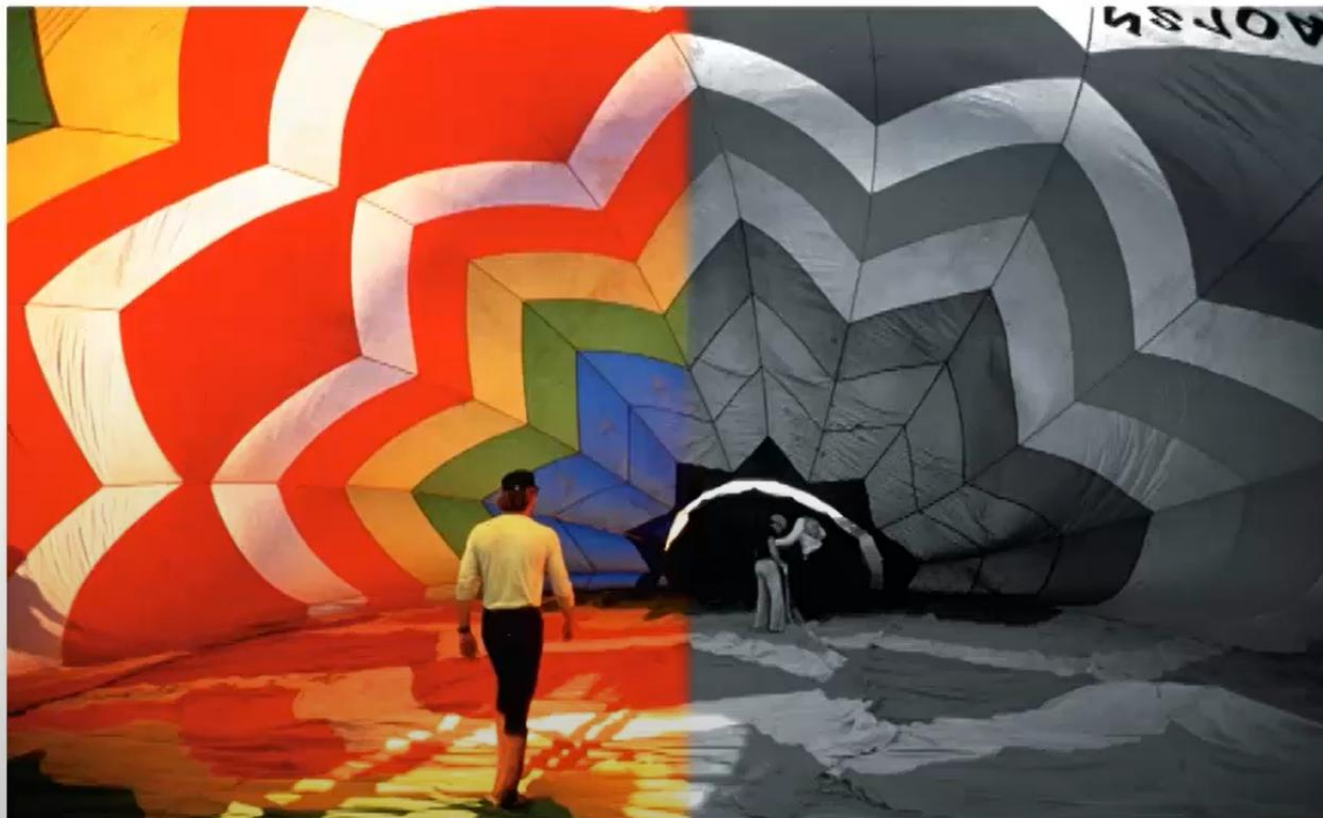
Deficits in visual perception

- Deficits in visual perception
 - Cortical blindness and “blindsight”
 - ✓ Lesions around **V1**
 - ✓ Hemianopia (half of visual field)
 - ✓ Quadrantanopia (1/4)
 - ✓ Scotoma (small region)



Deficits in visual perception

- Deficits in color perception
 - Achromatopsia (無色視覺)
 - ✓ “a”: without, “chroma”: hue
 - ✓ Lesions around **V4 (color)** area due to strokes and tumors

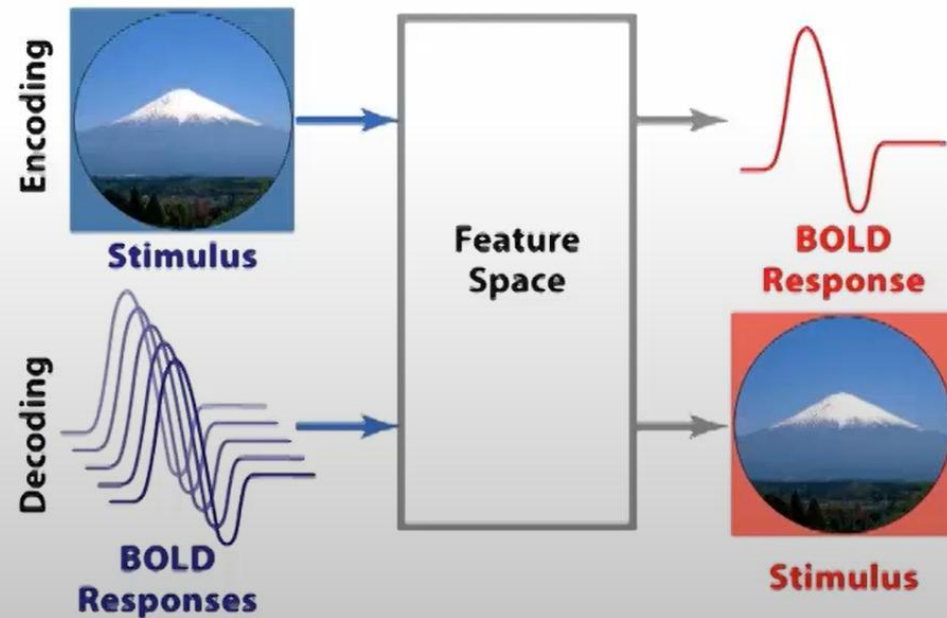
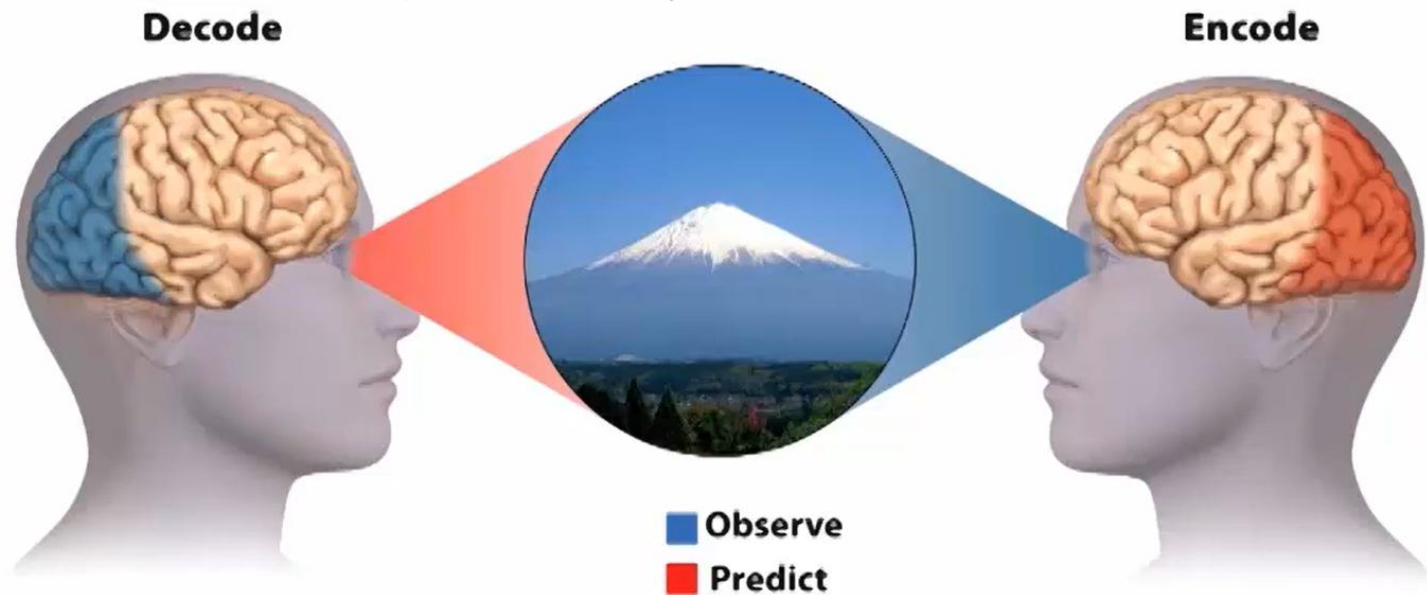


Deficits in visual perception

- Deficits in motion perception
 - Akinetopsia (motion blindness)
 - ✓ view the world as a series of snapshots
 - ✓ Deficits in **MT** area



Encoding vs. decoding



Different kinds of neural codes

- Rolls and Deco (2002) summarize three neural representations:
 - **Local representation** (= grandmother cells)
 - ✓ All the information about a stimulus/event is carried in one of the neurons.
 - **Fully distributed representation**
 - ✓ All the information about a stimulus/event is carried in all of the neurons of a given population.
 - **Sparse distributed representation**
 - ✓ A distributed representation in which a small proportion of the neurons carry information about a stimulus/event

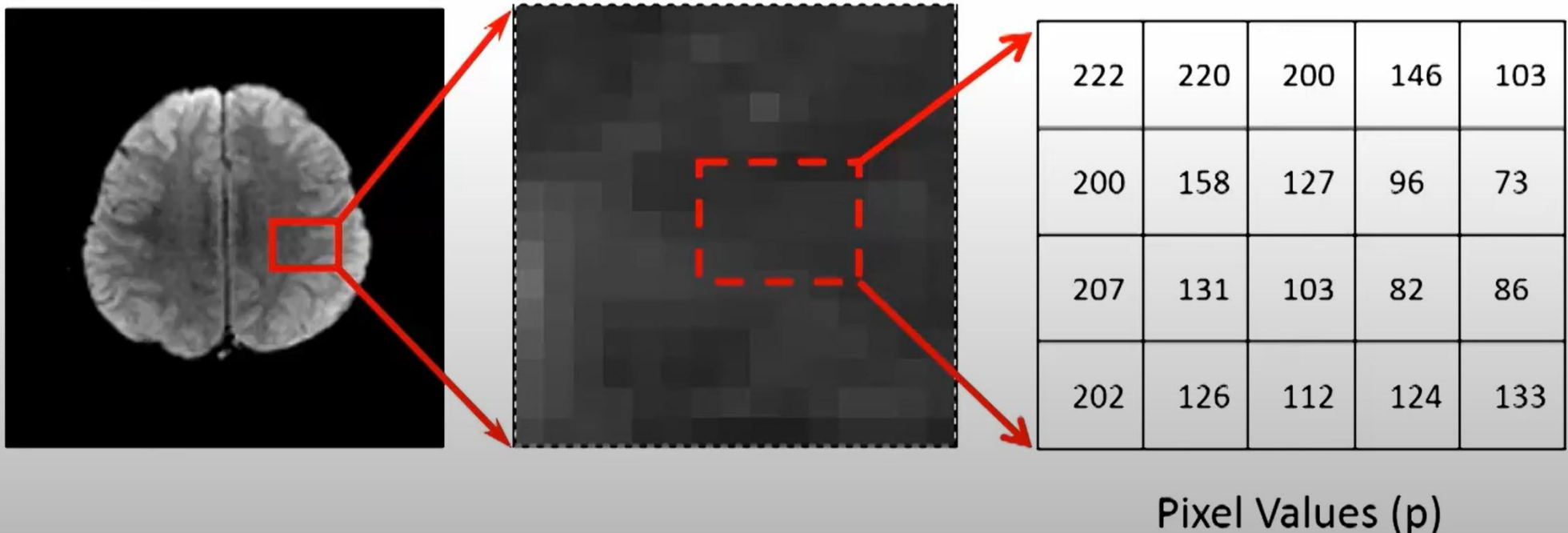


Advantages and disadvantages of ERP

- ERP signal is directly related to neural activity and this electrical activity is conducted instantaneously to the scalp
 - Therefore, ERP has an **excellent temporal resolution**
- ERP signal is derived from different sources in the brain and it is not possible to infer exactly where these sources are from the scalp
 - Therefore, ERP has a **poor spatial resolution**

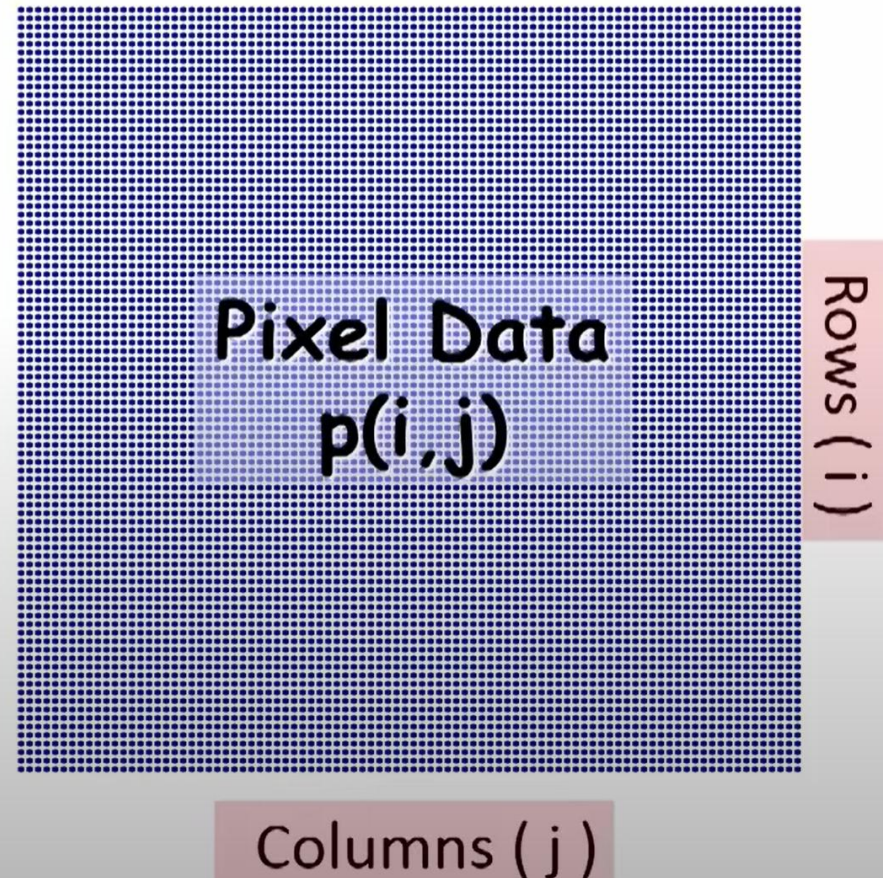
The basic concept of images

- What is the “format” of the image?
 - “digital” images
- A digital image = A matrix of numbers, w/ one number represents one pixel
- **Pixel** = Picture element
- A digital image = A mathematical matrix



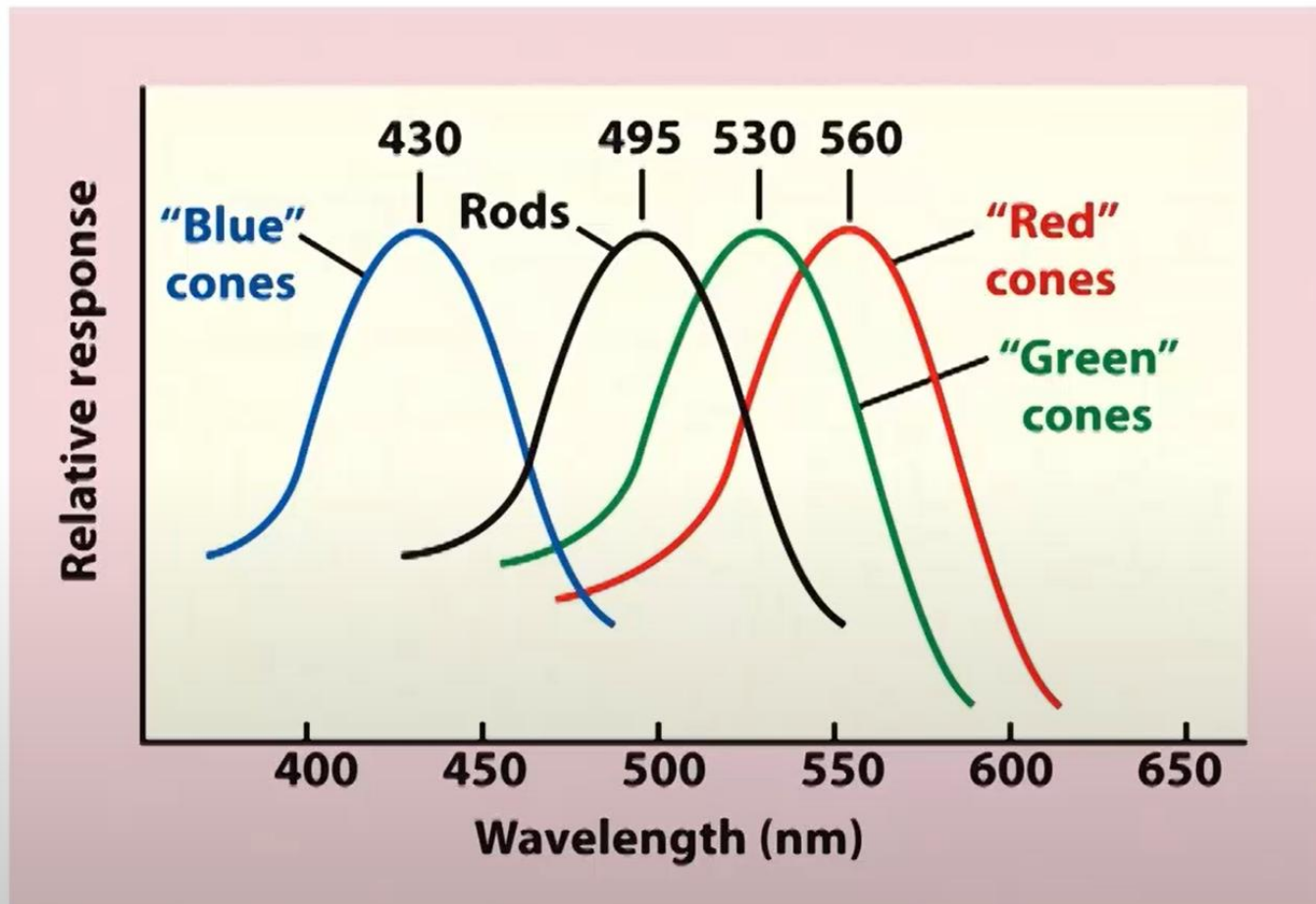
The basic concept of images

- Pixels and Image Matrices
- numbers of pixels = number of rows * number of columns



Retina and photoreceptors of the eye

- Two types of photoreceptors: rod and cone
- Spectral sensitivity functions for rods and 3 types of cones:



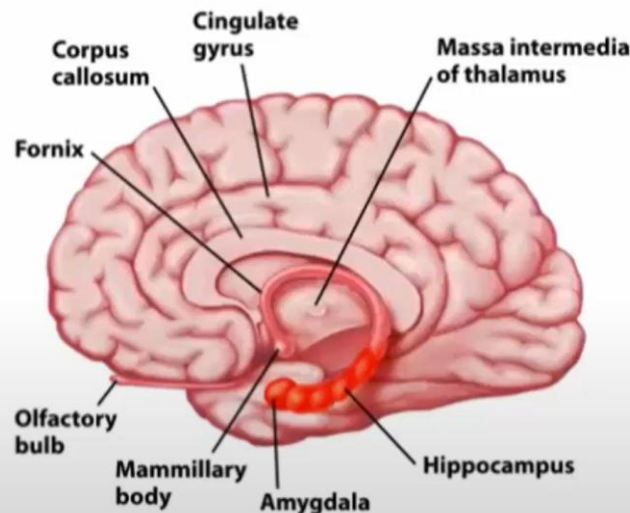
Different types of spatial representation

- Finding the nearest bank and picking up a pencil both require **spatial processing**
- Brain doesn't regard space as a continuous single entity
- Space in the brain exists in many forms:
 - Locations on sensory surfaces (e.g. retina; **retinocentric space**)
 - Location of objects relative to the body (**egocentric space**)
 - Location of objects relative to each other (**allocentric space**)



Different types of spatial representation

- Space
 - A common dimension of most perceptual systems and of motor/action system
 - e.g., a cat comes: by sight, sound, touch or smell.
- Cross-modal perception
 - Integrating information across sensory modalities



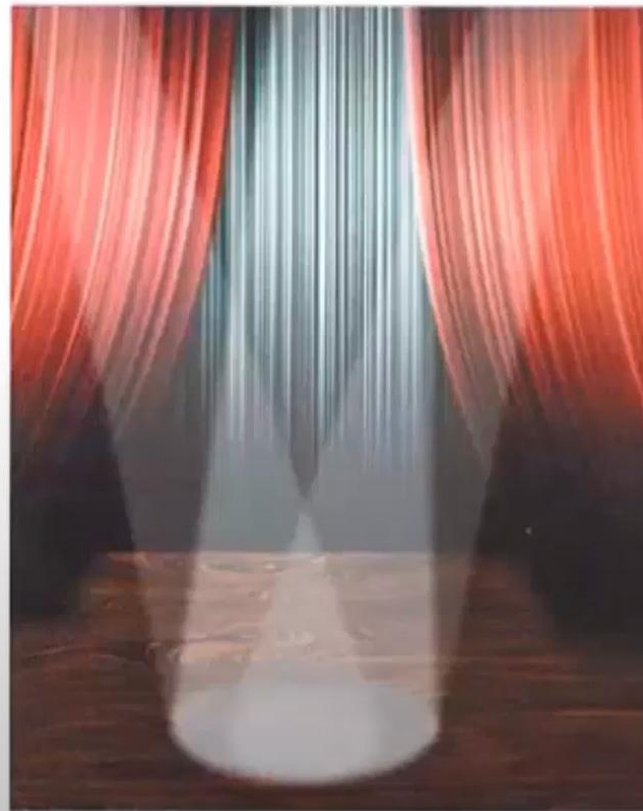
Space, attention and the parietal lobe

- Space and attention:
 - Attention tends to be **directed to locations in space** (space is a common dimension of different sensory systems and our motor system).
 - e.g., catch baseball
 - Attention may be **needed to bind together different aspects of conscious perception** (e.g. shape and color, sound and vision)
 - e.g., find faces and/or a vase



A “spotlight” on attention

- The spotlight:
 - highlights a particular location in space (a salient object)
 - move from one location to another (e.g., **eye movement**)

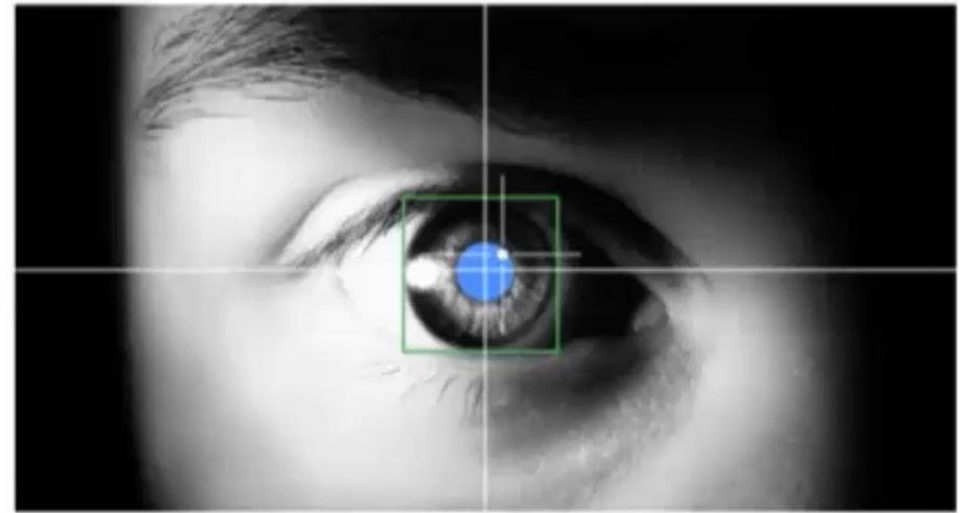


Eye movement: fixation location and saccade



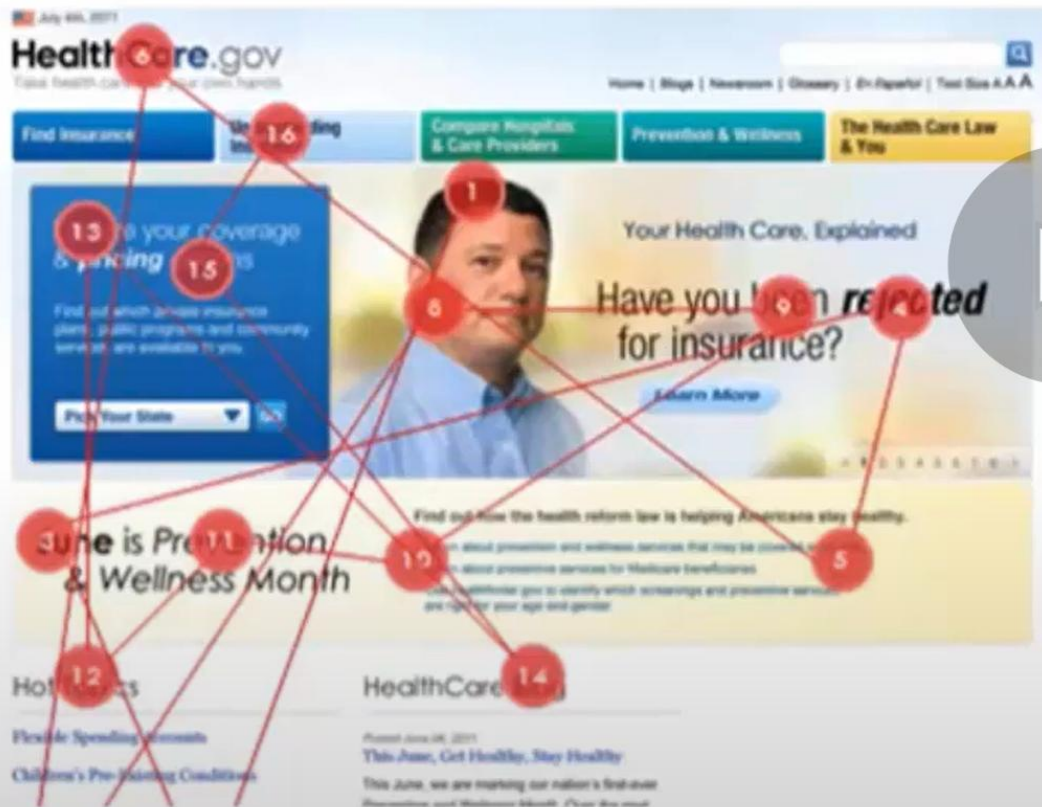
Eye-tracker

- ✓ Detecting pupil and corneal reflection
- ✓ Calibration: mapping eye movements to eye fixation positions

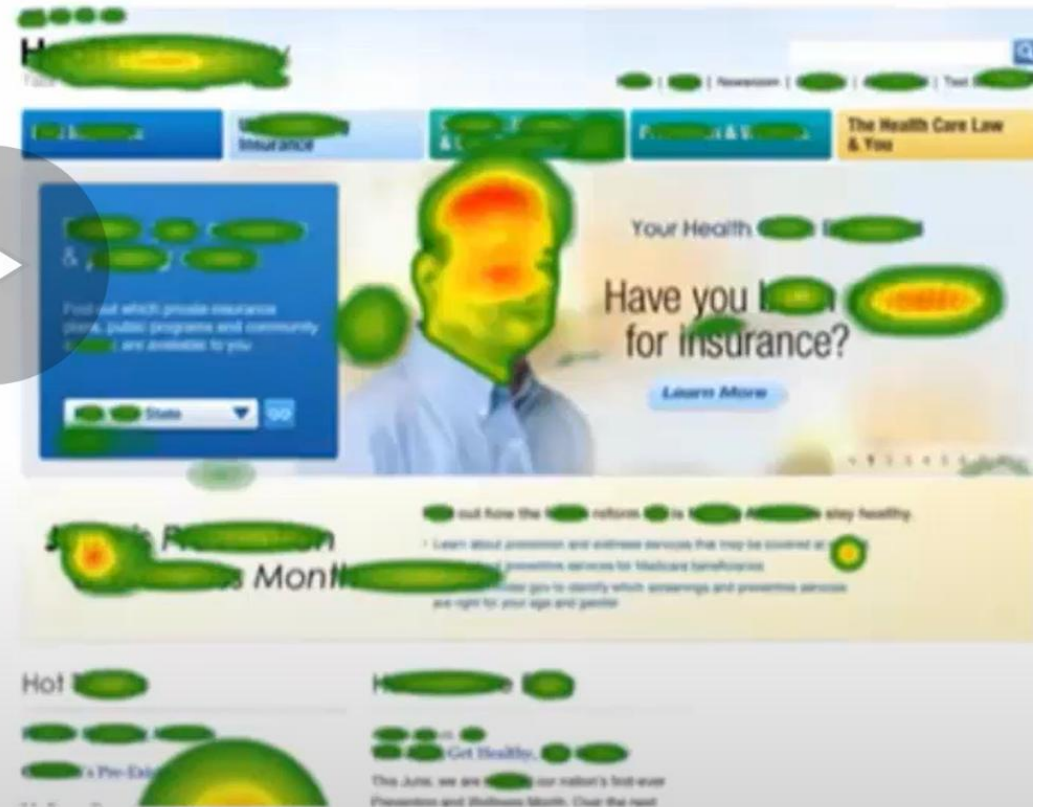


Eye-tracker

Eye movement map



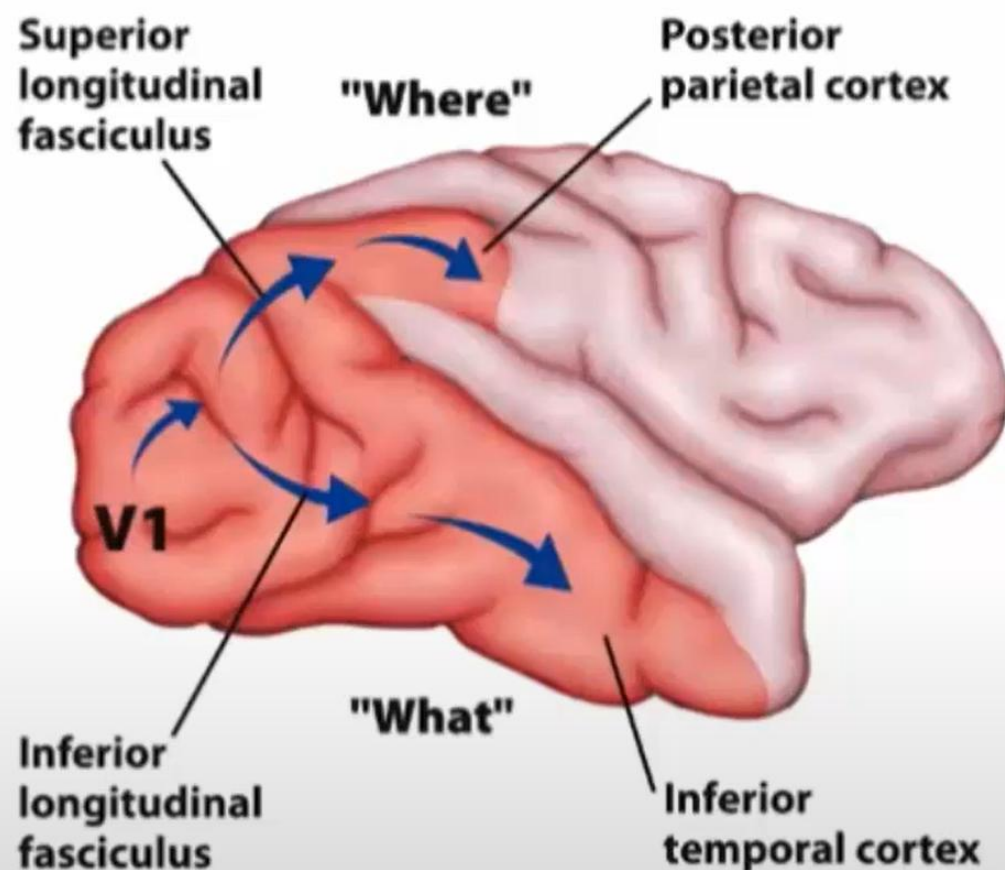
Fixation map



<http://is.gd/QZX5RF>

Two cortical pathways for visual perception

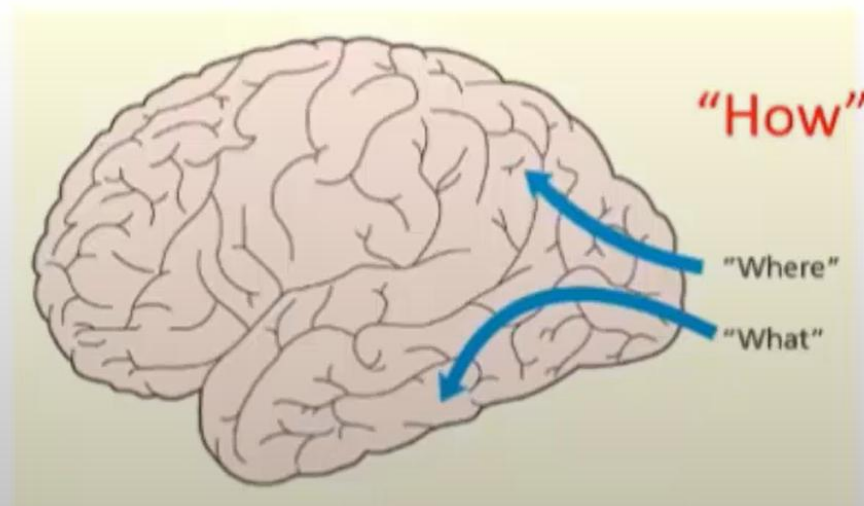
- Two major outputs from occipital lobe
 - **Dorsal pathway** leads to parietal cortex
 - ✓ “**where**” pathway
 - ✓ Spatial analysis
 - **Ventral pathway** leads to temporal cortex
 - ✓ “**What**” pathway
 - ✓ Object recognition



Space, attention and the parietal lobe

- **Parietal lobes:**

- Bring together different types of spatial representation that are needed for action (e.g. integrating visual space with body space), also called the "**how**" route
- Integrate visual spatial information with postural information -- essential for locating objects in space
 - ✓ motor cortex, auditory cortex, and visual cortex

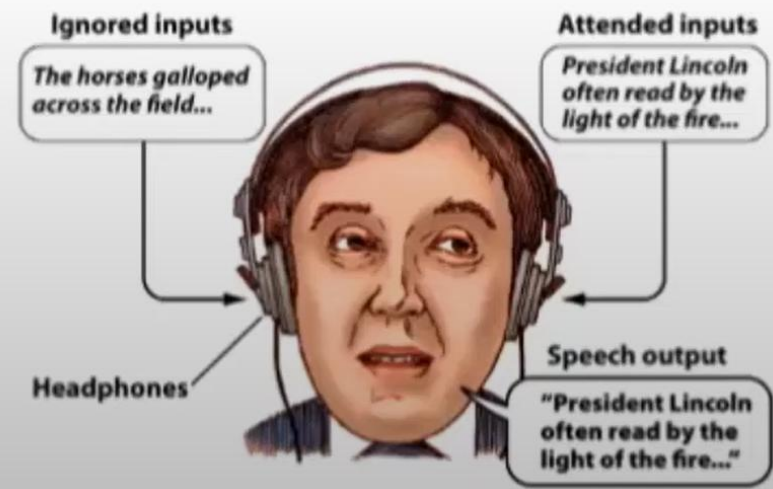
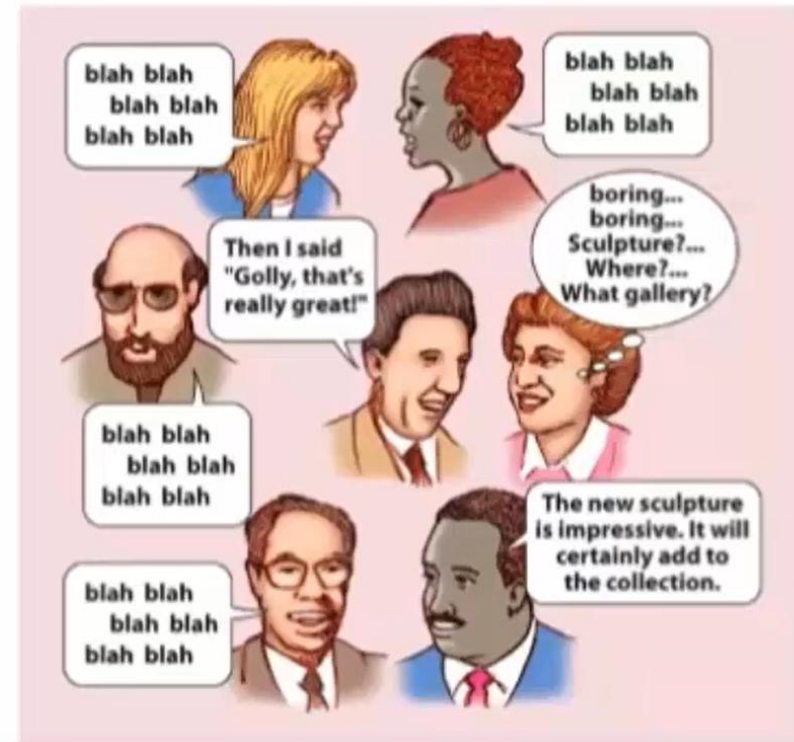


Attention

- Attention:
 - cognitive process by which certain information is **selected** for further processing and other information is discarded (as a filter or a bottleneck in processing)
 - ✓ Selection based on relevance/importance to current goals
 - ✓ Cocktail party effect

Cocktail party effect

- At a cocktail-party we can attend one conversation and ignore surrounding noise
- This is simulated in a lab through the shadowing paradigm
 - Dichotic listening
 - People can remember well only the attended conversation



- Pop-out search

➤ target can be identified by a single feature

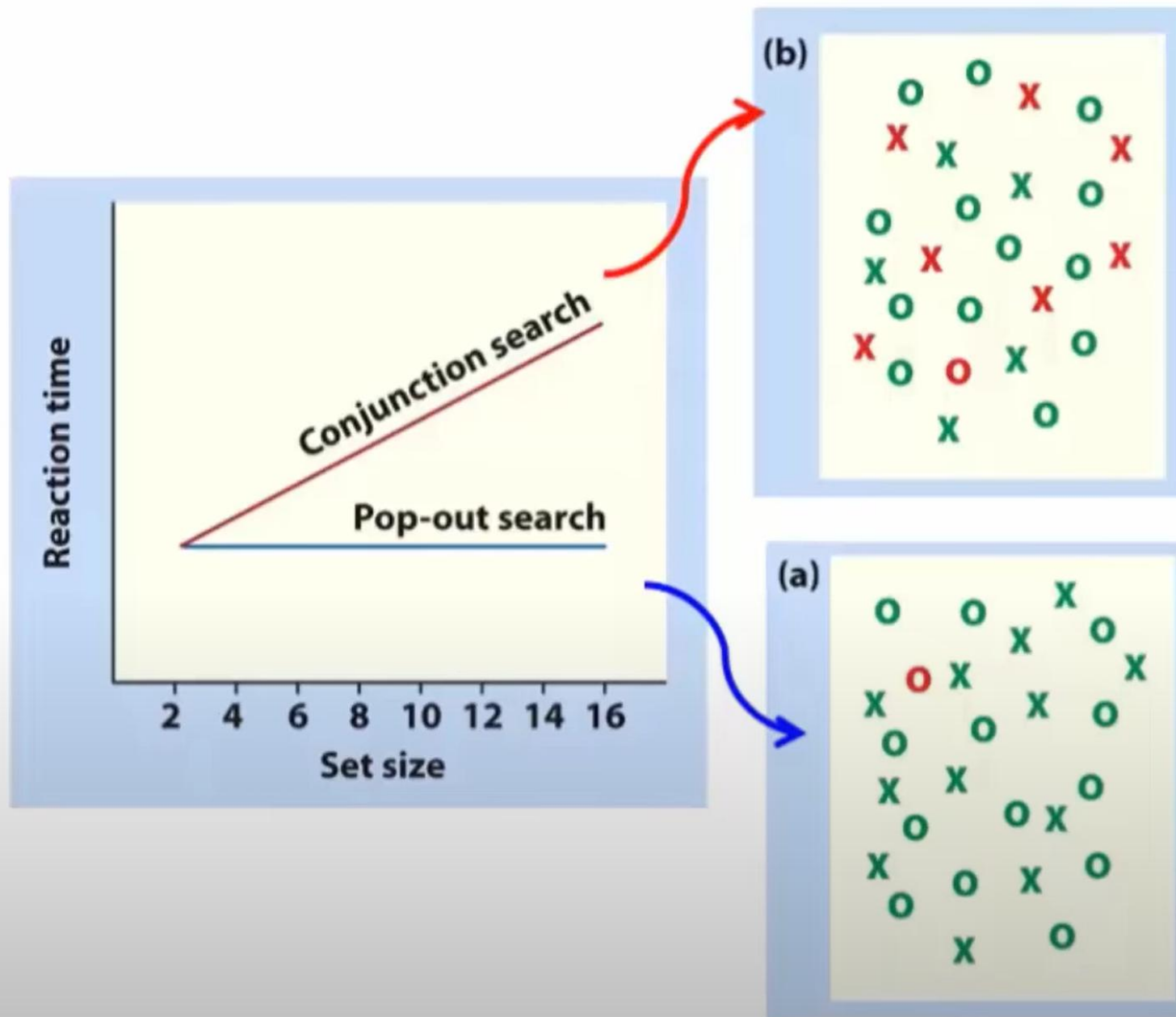
X		O	O	X		O		
	O	X	X		O		O	X
			O	X		X	X	
X	T	X		X		O	O	
O		O	X		O			
	X		O	O		O		X
O				X		O		X

- Conjunction search

- the target is defined by the conjunction of two or more stimulus features (e.g., the color red and the letter T)

XX	O	X	O	O	XOO	X	O
O	X	X	OXO	OX		O	X
	O	XO	X	O	X	XX	
X	O	X		XO		OO	O
O	X	XO	X			O	XX
	X	X	OX	XO		OO	X
O				XO	O	OX	
X		X		OX	O	X	O
OO	XO		XOXO	OO	XO	XX	

Pop-out search and conjunction search



Computational issues in motor controls

- **Hierarchical representation** of action sequences
 - High-level representation of the action is independent of any particular muscle group.
 - Motor representations are not linked to particular effector systems

